GRANT NEILSON

Engineering focused Computational Biologist with seven years of industry experience building scalable and robust analysis pipelines, developing internal Python & R packages and processing large-scale whole genome and single cell sequencing datasets (500GB+). Committed to writing production-quality code through rigorous testing, version control, and thorough documentation. Active open-source contributor, passionate about developing reliable scientific software and advancing reproducible research through community-driven collaboration.

INDUSTRY EXPERIENCE

2026 | 2023

Senior Computational Biologist

CoSyne Therapeutics — London, UK

- Led analysis of large-scale genomic datasets to support biomarker discovery and target id, translating complex results into clear, decision-ready outputs for cross-functional teams. Culminated in the successful partnership with a publicly traded biotech.
- Co-Developed and maintained internal & open source R and Python packages. For statistical modeling, data processing and visualisation. Bixverse, is a Rust based R package enabling the analysis of 3 million single cells on a standard laptop.
- Developed internal container system using Docker and ECR, to enable reproducible research. This system used CI/CD to build images robustly and securely.
- Recognized as a named contributor in version 3.4.1 of the NF-core Sarek pipeline. Optimised the running of the pipeline, achieving a 50% cost reduction and 30% decrease in computational time per sample.

2023 | 2020

Computational Biologist

Fios Genomics — Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Conducted complex regression model analyses for biomarker discovery in bladder and lung cancer.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

View this CV online

CONTACT

- grant.neilson5@gmail.com
- GitHub
- LinkedIn

LANGUAGES

- R
- Python
- Bash
- Nextflow

TECHNOLOGIES

- GitLab/GitHub
- Docker
- aws
- Azure

OPEN SOURCE & PUBLIC PROJECTS

- bixverse
- mainfoldsR
- cosyne-cnvutils
- Sarek

Made with the R package pagedown.

The source code is available on GitHub.

Last updated February 2026.

2020 | 2019

Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School — Exeter, UK

- Developed novel bioinformatics pipelines for Epigenome/Genome wide association studies, mQTL analysis and quality control of Illumina EPIC array data.
- Took the lead on a project performing EWAS and GWAS analysis in ADHD patients. This involved fitting linear and logistic regression models while accounting for covariates.
- Was selected for a secondment to New York to work at the Mount Sinai medical school as a visiting bioinformatician. This was unfortunately cancelled due to COVID-19.

EDUCATION

2019 | 2015

MSCi Biology

University of Southampton — Southampton, UK

- **Grade** : First Class Honors
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015 | 2013

School

Merchiston Castle Shool — Edinburgh, UK

- A Levels: Biology, Chemistry, Maths

SELECTED PUBLICATIONS

2025

Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, **Grant Neilson**, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preproccessing
- doi: https://doi.org/10.1101/2025.04.17.649387

2025

Detecting cell level transcriptomic changes of Perturb seq using Contrastive Fine tuning of Single Cell Foundation Models

BioRxiv

- Wenmin Zhao, AnaSolaguren-Beascoa, **Grant Neilson**, Regina Reynolds, Louwai Muhammed, Liisi Laaniste, Sera Aylin Cakiroglu
- Role: Data Preproccessing
- doi: https://doi.org/10.1101/2025.04.17.649395